

Ariel Lubonja

PhD Candidate, Scalable & Distributed Machine Learning

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EDUCATION

Johns Hopkins University

PhD, Computer Science — GPA: 3.92/4.0

Aug 2022 – May 2028 (Exp.)

- Advisors: Randal Burns, Joshua Vogelstein, Alan Yuille

MS, Data Science — GPA: 3.8/4.0 — Dean's List

Jan 2021 – May 2022

RESEARCH EXPERIENCE

Yggdrasil Decision Forests (Google) — Open-Source Research Collaboration May 2025 – Present

Joint work with the Google YDF team (Zurich)

- Reduced Google YDF's sparse oblique training time by $\sim 6\times$ (12 h $\rightarrow$ 2 h) on high-dimensional data via 7 upstream pull requests (3 merged, 4 in review).
- Designed two further methods adding a $1.7\text{--}2.5\times$ end-to-end speedup over this baseline (arXiv preprint): SIMD-vectorized histogram binning (two vector compares replace per-sample binary search) and a per-node adaptive splitter that picks exact vs. histogram splits by cardinality.
- Extending this work to multi-node distributed training and SIMD-vectorized inference via QuickScorer (ongoing).
- Mentored 2 graduate students (one MS, one junior PhD).

NVIDIA PhysicsNeMo Contributor

Nov 2025 – Feb 2026

2 merged pull requests

- Implemented and tested a new API in NVIDIA PhysicsNeMo; merged upstream in 2 pull requests (1, 2).
- Cut runtime by 60% on low-resolution inputs by eliminating a redundant call.
- Funded by and contributing to the Johns Hopkins Turbulence Database (JHTDB), the largest public turbulence-simulation archive and a PhysicsNeMo data source.

Scalable Node Embedding for Graph ML

Dec 2023 – May 2024

Open-source code and IPDPS 2024 workshop paper

- Cut node embedding of the 1.8B-edge Friendster graph from ~ 56 min to 6.4 s on 24 cores (500 $\times$; 17 $\times$ vs. a Numba JIT baseline) via a lock-free, edge-parallel Ligra reformulation.
- Published the edge-parallel Graph Encoder Embedding (GEE) at the IPDPS 2024 Workshops (GrAPL); single-pass GEE is an order of magnitude faster than spectral, node2vec, or GCN methods even before parallelization.

GPU-Accelerated Orthogonal Matching Pursuit

May 2022 – Jan 2023

Python package (PyPI) and arXiv preprint

- Built a drop-in GPU replacement for scikit-learn's Orthogonal Matching Pursuit (OMP), up to 310 $\times$ faster.
- Applied Gramian-matrix precomputation and Cholesky factorization for up to 83 $\times$ faster single-core execution.
- Implemented batching for CPU and GPU acceleration of large matrices using cuBLAS.
- Released as PyPI package [batched-omp](#), with an [arXiv preprint](#) and [open-source code](#).

WORK EXPERIENCE

Software Developer

May 2017 – Jan 2021

jBoxers

Sofia, Bulgaria

- Built an [Alice in Wonderland chatbot](#) for the University of Southern California using Mycroft open-source speech-to-text tools, a custom NLP model, and Google Dialogflow.
- Developed frontend (Angular) and backend (Django, Java EE) for the Library of Congress's [Virtual Card Catalog](#).
- Supervised 2 summer interns.

TECHNICAL SKILLS

Languages & Tools: C++, Python, CUDA, Git, Bazel

HPC & Performance: MPI, Slurm, SIMD (AVX-512, Google Highway), GPU profiling (Nsight Systems), cuBLAS, Dask, Zarr

Machine Learning & Mathematics: PyTorch, Tree Ensembles, Graph Learning, Sparse ML, Numerical Optimization

HONORS & LEADERSHIP

Fellowship to Johns Hopkins University: Fulbright-style fully funded national fellowship — 1 of 14 nationwide

Vice President: Upsilon Pi Epsilon (international CS honor society), Johns Hopkins chapter